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Avadi, Chennai

Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 25-03-2026

Full-Stack Skill Learning Academy Marketplace with Secure Authentication

SDG: SDG 4 – Quality Education

Presented By:

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Courses Handling Faculty:

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INTRODUCTION



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- With the rapid growth of online education, there is a need for platforms where learners can easily discover and enroll in skill-based courses.
- With the rapid growth of online education, there is a need for platforms where learners can easily discover and enroll in skill-based courses.
- This project aims to solve that problem by creating a **Full-Stack Skill Learning Academy Marketplace**.
- The platform allows instructors to publish courses and students to browse, search, and enroll in them.
- Secure authentication and **role-based access control** ensure safe access for Admin, Instructor, and Student users.
- The system helps instructors manage their courses while students can track their enrollments and learning progress.
- Admin users can monitor platform activity, manage users, and view analytics.
- Overall, the system demonstrates a real-world **full-stack web application using HTML, CSS, JavaScript, Node.js, and MySQL**.

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PROBLEM STATEMENT



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- Many online learning platforms do not provide a simple marketplace where instructors can easily publish courses and students can discover them in one place.

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- Existing systems often lack proper **role-based access control** for Admin, Instructor, and Student users.

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- Managing courses, enrollments, and user data manually or through separate systems leads to inefficiency.

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- Students also face difficulty in searching and enrolling in skill-based courses from different sources.

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- There is a need for a centralized platform that connects instructors and learners efficiently.

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-The system should provide **secure authentication** and proper user management.

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- It should also allow instructors to manage courses and students to track their learning progress.

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- Therefore, a **Full-Stack Skill Learning Academy Marketplace** is required to provide an organized and automated learning platform.

OBJECTIVE(S)



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- To design and develop a **Full-Stack Skill Learning Academy Marketplace** with secure authentication and role-based access for Admin, Instructor, and Student users.
- To implement a **course management and enrollment system** that allows instructors to create courses and students to browse, enroll, and track learning progress.

PROPOSED SYSTEM



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- The system provides a **web-based marketplace** where students can browse and enroll in skill-based courses.

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- It supports **secure user authentication** with role-based access for Admin, Instructor, and Student.

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- Instructors can **create, update, and manage courses** through their dashboard.

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- Students can **search courses, view details, enroll, and track their progress.**

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- Admin can **monitor users, courses, and platform statistics** through an admin dashboard.

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- The backend uses **Node.js (Express)** to handle API requests and business logic.

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- The system uses **MySQL database via XAMPP** to store users, courses, enrollments, and payment data.

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- The frontend built with **HTML, CSS, and JavaScript** provides a responsive

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interface for all users.

PROPOSED SYSTEM ARCHITECTURE



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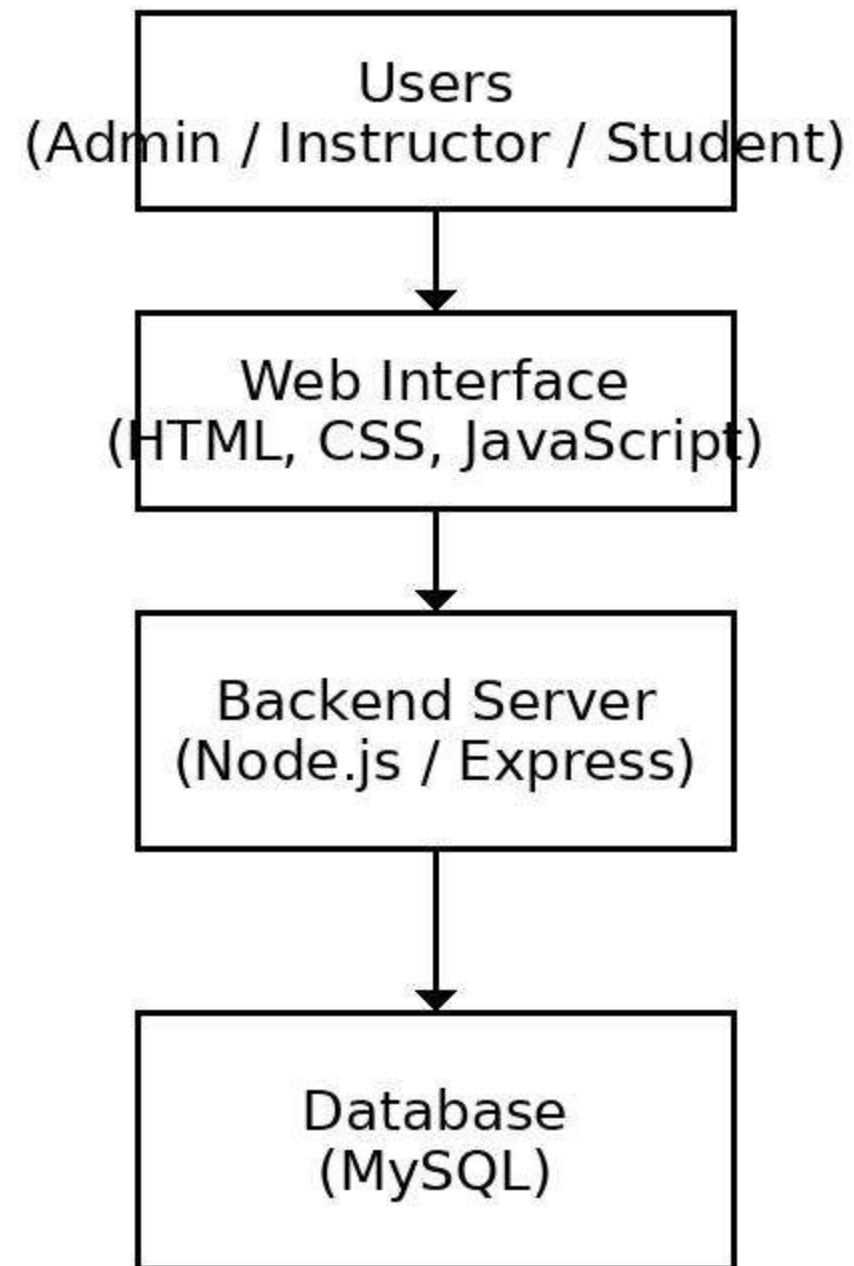
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WORKFLOW



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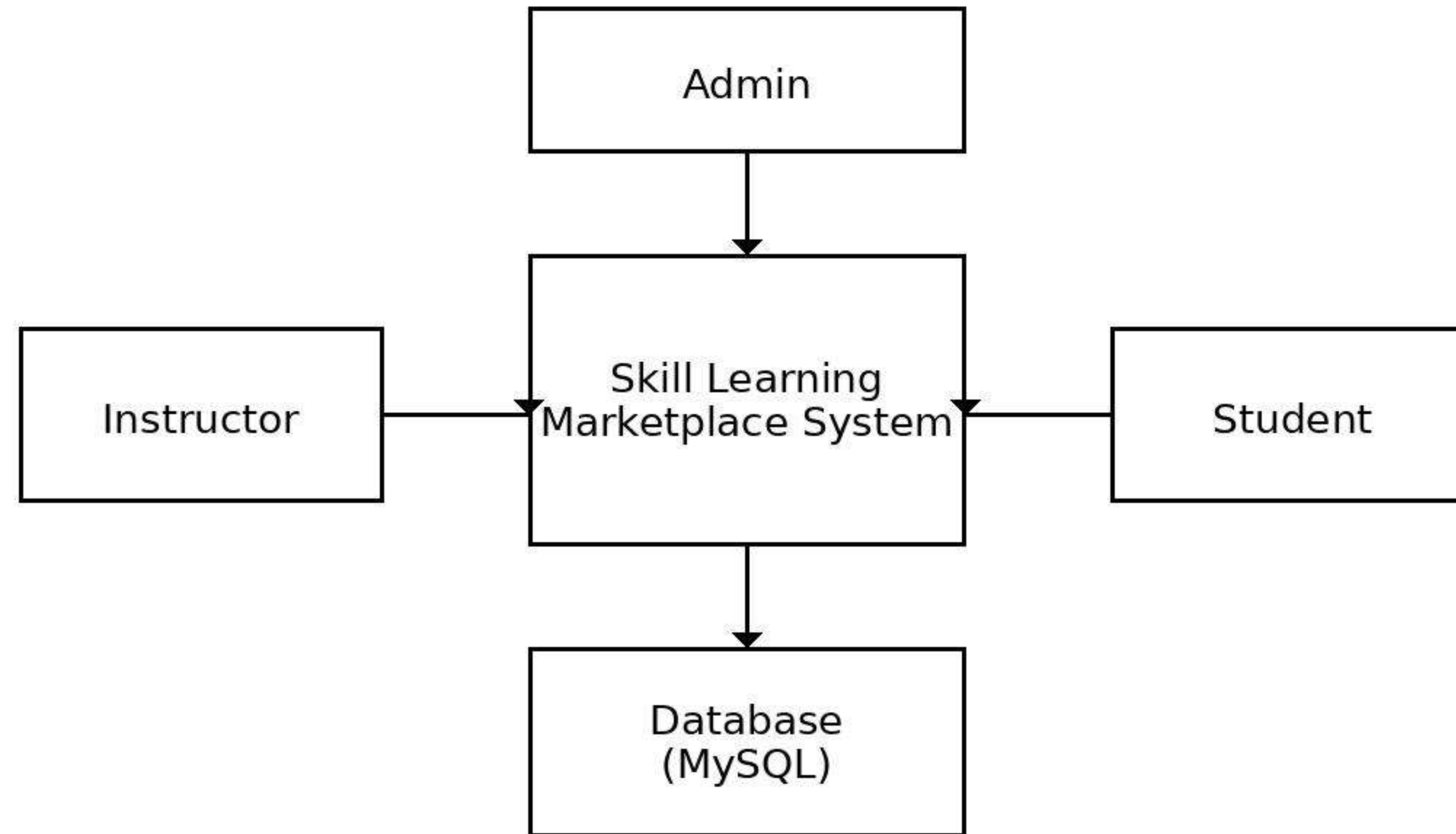
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TECHNOLOGIES USED



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Frontend:

HTML5, CSS3, JavaScript (ES6)

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Backend:

Node.js with Express.js

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Database:

MySQL (using XAMPP)

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Tools:

Visual Studio Code, XAMPP, Git/GitHub, Postman, Web Browser (Chrome)

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Hardware/Software Requirements:

Processor: Intel i3 or higher

RAM: Minimum 4 GB

Operating System: Windows/Linux/MacOS

Software: Node.js, XAMPP, Web Browser (Chrome/Edge)

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MODULE DESCRIPTION



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Module 1: User Authentication & Management

Handles user registration, login, logout, and profile management with secure authentication and role-based access (Admin, Instructor, Student).

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Module 2: Course Management

Allows instructors to create, update, delete, and manage courses, including course details and categories.

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Module 3: Enrollment Management

Enables students to browse courses, enroll in selected courses, and track their learning progress.

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Module 4: Admin Management & Analytics

Allows the admin to manage users, monitor courses and enrollments, and view platform statistics and activity reports.

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IMPLEMENTATION SCREENSHOTS



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TrackWise

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STUDENT

MAIN MENU

Dashboard

Courses

Learning Paths

ACCOUNT

My Profile

Achievements

Sign Out

Active courses

Enrolled

Courses done

Completed

Currently learning

In Progress

Badges earned

Achievements

My Courses

No courses yet

Browse our catalog and enroll in your first course!

Explore Courses

Achievements

Complete courses to earn achievements!

Course Progress

IMPLEMENTATION SCREENSHOTS



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Learn Skills That Actually Matter

Join thousands of professionals leveling up with expert-led courses, real projects, and verified certifications.



Structured Learning Paths
Curated skill tracks from beginner to expert



Verified Certificates
Industry-recognized credentials on completion



Multi-Role Ecosystem
Students, Instructors & Admins — one platform

Welcome back

New here? [Create your account](#) →

⚡ DEMO QUICK LOGIN

ADMIN
admin@trackwise.com

INSTRUCTOR
john@trackwise.com

STUDENT
alice@example.com

or enter credentials manually

EMAIL ADDRESS

pallavi123

PASSWORD

.....

☐ Remember me for 30 days

[Forgot password?](#)

Sign In Securely →

🔒 Protected by rate limiting, encrypted sessions & bcrypt hashing

IMPLEMENTATION SCREENSHOTS



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STUDENT

MAIN MENU

- Dashboard
- Courses
- Learning Paths

ACCOUNT

- My Profile
- Achievements
- Sign Out

Course Catalog

Explore all available skills and learning paths

All Categories

All Levels

All

Blockchain

Cloud & DevOps

Cybersecurity

Data Science

Database

Mobile Development

UI/UX Design

Web Development

INTERMEDIATE

★ 4.90

WEB DEVELOPMENT

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From zero to hero in JavaScript. Covers ES6+, async/await, DOM manipulation and ...

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INTERMEDIATE

★ 4.60

CLOUD & DEVOPS

AWS Cloud Practitioner

Complete AWS certification prep with hands-on labs and real-world projects...

IMPLEMENTATION SCREENSHOTS



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TrackWise

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STUDENT

MAIN MENU

Dashboard

Courses

Learning Paths

ACCOUNT

My Profile

Achievements

Sign Out

Welcome, pallavi! 🙌

Continue your learning journey

Active courses

1

Enrolled

Courses done

0

Completed

Currently learning

0

In Progress

Badges earned

0

Achievements

My Courses

BEGINNER

DATA SCIENCE

Python for Data Science

🕒 38h 📖 155 lessons ⭐ 4.70

0% complete

Achievements

🏆

Complete courses to earn achievements!

🔔

⚙️

+ Explore Courses

ADVANTAGES & LIMITATIONS



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Advantages

- Provides a centralized online learning platform.
- Supports secure login with role-based access.
- Easy course creation and enrollment.
- Efficient data management using MySQL.
- Improves accessibility to skill-based courses.

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Limitations

- Requires internet connection.
- Setup may be difficult for beginners.
- Limited advanced features in basic version.
- Security risks if not properly maintained.
- Performance may reduce with many users.

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Thank you

